

Design and Performance Analysis of Functionalized Carbon Nanotube (CNT)-Natural Rubber Composites for Aircraft Landing Gear Shock Disk Applications

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Abstract:

Design and performance analysis of functionalized carbon nanotube (CNT)-natural rubber composites for applications in aircraft landing gear shock disk are the aims of this research. Mechanical properties of the fabricated composites are investigated in this research, and the suitability of these composites to be used under high-stress conditions is discussed. Selected for their elasticity and strength, the CNTs were functionalized in order to enhance their dispersion within the matrix—a natural rubber and polymer material. Tensile strength testing and compression were among the mechanical tests used in an orderly experimental approach using the preparation of composite samples. It shows that the presence of functionalized CNTs significantly increases the mechanical strength, toughness, and impact resistance of the composites and therefore puts them as favorable candidates to find application in the aerospace sector. The studies are significant on the ground that they could be of great benefit to the aerospace sector to devise lightweight but high-performance material.

Nomenclature:

CNT	Carbon Nanotubes	ΔL	Change in length
NR	Natural Rubber	L_0	Initial length
EPDM	Ethylene Propylene Diene Monomer	F_a	Compressive force
MWCNT	Multi-Walled Carbon Nanotubes	A	Cross-sectional area
UTM	Universal Testing Machine	F	Applied force
DMA	Dynamic Mechanical Analysis	σ_a	Compressive Stress

Introduction:

Increased aerospace demand for composite materials of high performance has encouraged the development of new concepts in enhancing critical components' performance at severe stress levels. Composites of carbon nanotube (CNT) with natural rubber have been found as a limelight material through their twin attribute of mechanical strength coupled with enhanced elasticity. CNT functionalization leads to better dispersion in the matrix of polymer, and thereby considerably the total property of the material. This study examines the possibility of functionalized CNT-natural rubber composites being used in aerospace applications, such as landing gear shock disks, where strength and impact resistance are critical.

Natural rubber is well known to have the ability to withstand impact, being robust and having high energy absorption and dispersion capability in an effective manner. This has made it to be utilized mainly in cases that demand shock absorption, for example, shock disks utilized in landing gears. Natural rubber is thus widely utilized in the aerospace industry in parts endured repeated mechanical load [1]. Natural rubber's elasticity is convenient but not so the mechanical strength to endure high demanding applications, and that is where CNTs are utilized.

CNTs have superior mechanical characteristics such as high tensile strength, stiffness, and thermal conductivity. They are most useful when utilized in polymer matrices, e.g., natural rubber, because they improve the mechanical properties of the composite material without contributing to excessive weight. CNTs reinforce the mechanical properties, toughness, and resistance to wear of the composite, and therefore they are highly sought after to be applied in aerospace engineering. CNTs addition to rubber improves these properties in addition to increasing the energy dissipation ability of the composite, a very important property in landing gears' shock discs required to endure heavy impacts during the take-off and landing processes [2]. One of the challenges of using CNTs in polymer matrices is that CNTs agglomerate, thus uniform dispersion is not favorable for optimal material performance. Functionalization, chemical modification of CNT surfaces, addresses the problem by enhancing CNT-polymer matrix affinity and improving even dispersion. The process greatly improves the mechanical, electrical, and thermal qualities of the composite, making functionalized CNT-natural rubber composites more efficient and durable under high-stress environments [3].

CNT uses in composites intended for aviation applications have been an area of considerable study.

It has been found through previous studies that functionalized CNTs improve material electrical and mechanical properties utilized in aircraft structures. Functionalized CNTs, for instance, have been proven to improve epoxy adhesives' impact resistance and durability, which are vital in aviation applications [4]. The capacity to customize the dispersion of CNTs in the composite matrix allows the material to be capable of fulfilling the rigorous requirements of aerospace components, such as those with high-impact loads like shock disks for landing gear. With the aerospace industry still demanding lighter, performance-critical materials, the CNT-natural rubber composite is the ideal solution. They exhibit enhanced mechanical properties like increased strength and toughness, which are very important in components requiring repeatable operation under conditions of high stress. The use of CNTs also permits increased energy dissipation to enhance shock absorbing capability in a material required in applications such as landing gear shock disks [5][6]

Overall, functionalization of CNTs and their employment in natural rubber composites form a cutting-edge technology for application in the aerospace sector. The composites contain increased mechanical toughness, shock, impact resistance, and are well-suited to sensitive aerospace components. This work demonstrates evidence of the ability of CNT-natural rubber composites to cater to the shifting demands of the aerospace industry toward lightweight, high-performance, long-life materials.

State of the Art:

CNT-rubber composites have shown improved mechanical stability during thermal aging. CNT is reported through experiments to improve tensile strength, elongation, and anti-aging strength and can be applied for aerospace components in a changing temperature environment [7].

Manufacturing procedures like melt blending, solution casting, and in-situ polymerization are crucial in the end dispersion of CNTs in rubber matrices. Efficient dispersion procedures improve mechanical and thermal properties by reducing agglomeration issues during the synthesis of nanocomposites [8].

Functionalization methods coat reactive functionalities onto the CNT surface, which enhance matrix polymer compatibility. Enhanced dispersion, interfacial bond strength, and load transfer follow, ultimately causing enhanced mechanical properties of the composites [9].

Carbon fiber reinforced hybrid composites have higher mechanical strength, impact resistance, and durability, properties of critical importance for aerospace structures subjected to dynamic loading conditions [10].

The incorporation of multi-walled CNTs into ethylene propylene diene rubber (EPDM) has been demonstrated to exhibit increased strain sensitivity along with mechanical properties. The materials are multifunctional with mechanical strength as well as electrical sensitivity, useful for sensing and structural integrity applications [11].

Current studies and molecular dynamics also suggest that the most significant factors controlling the mechanical behavior of rubber composites are CNT alignment, dispersion, and aspect ratio. In alignment with each other and with uniform dispersal, CNTs are found to cause tremendous improvements in toughness, strength, and stiffness [12].

Molecular dynamics simulations show that functionalization and defect levels in CNTs are key to rubber composite mechanical properties and creep strength. Well-functionalized CNTs increase adhesion and stress transfer but can increase flexibility or reduce mechanical strength depending on defect distribution [13].

Surface-functionalized CNTs in polystyrene matrices exhibited enhanced mechanical, electrical, and thermal properties, findings of interest to rubber industry composites, reflecting the influence of surface modification on the optimization of composite performance [14].

It is important to have uniform dispersion and good interfacial adhesion by adopting proper functionalization techniques in order to realize full benefit of the composites with CNTs as reinforcement phase. Agglomeration is addressed by using traditional surface modification techniques like acid oxidation and polymer grafting [15].

Specific analyses recognize the tensile strength, conductivity, and thermal stability enhancement in polymer composites reinforced with CNTs and thus gaining importance as materials to be used in aerospace components with light weight and extreme durability as the primary demands [16].

Mechanical evaluation is always seen with pertinent enhancements in tensile strength, stiffness, and impact resistance of CNT-reinforced composites ensuring their use under high-load and high-strain conditions in aerospace applications [17].

CNTs-based rubber nanocomposites have high thermal conductivity and mechanical strength. The above improvement is strongly required in the aerospace sector to withstand alternating thermal and mechanical loading [18].

Experiments confirm that addition of CNTs in natural rubber and ethylene-propylene-diene rubber matrices improves tensile strength, increases modulus, and improves fatigue life, thereby making the above composites capable of working under severe operating conditions [19].

Modeling methods, such as analytical and multiscale simulations, provide valuable predictions of the mechanical properties of CNT-reinforced composites. These methods are useful to customize CNT orientation and dispersion to obtain desired mechanical performance in loading conditions [20].

Thermal and mechanical performance study shows that the incorporation of CNT enhances not only mechanical but also thermal dissipation abilities, a double advantage for materials experiencing mechanical and thermal cycling [21].

Synergistic hybrid filler effects of carbon nanotubes and carbon black in natural rubber composites are found to result in improved mechanical properties and higher thermal conductivity, suggesting an open flexible design window for optimal aerospace material [22].

Molecular dynamics simulations demonstrate that vulcanized natural rubber composites filled with carbon nanotubes exhibit improved tribological properties such as low friction and wear resistance. These are required in continuous dynamic stress aerospace components [23].

Methodology:

This chapter encapsulates the carbon nanotube (CNT)-reinforced natural rubber composites' preparation process to improve natural rubber's mechanical and thermal properties. These properties are of interest in their use in aerospace industries, for example, shock disks for landing gears in airplanes. The objective of this process is to obtain homogeneous dispersion of CNTs in the rubber matrix, followed by intensive mechanical testing for obtaining the composite's performance for tensile as well as compressive loading.

1. Materials Utilized:

The following materials were utilized in processing the CNT-natural rubber composite:

Natural Rubber (NR): As NR has high elasticity, good energy absorption, and high impact resistance, it is the best material that can be used to absorb shock and hence would have the best utilization in aerospace applications.

Carbon Nanotubes (CNTs): CNTs were used as they possess extremely good mechanical stiffness, strength, and thermal conductivity, thereby improving the composite properties to a large extent.

Zinc Oxide (ZnO): Used as vulcanization accelerator, assisting in vulcanization acceleration and further improvement in strengthening properties of rubber.

Sulfur: Used as crosslinking agent when used in vulcanization and produces additional strength and toughness in the developed composite.

CBS (N-cyclohexyl-2-benzothiazylsulfenamide): Accelerator-type assisting in curing reaction strengthening as well as proper setting of rubber.

Stearic Acid: Aids in the uniform dispersion of CNTs and other fillers within the rubber matrix.

2. Preparation of Composite:

Optimal dispersion of CNTs within the rubber matrix is an organized procedure. Following is a step-by-step description of the procedure adopted to prepare the composite:

2.1. Drying of Ingredients:

To prevent early response or incompatibility on mixing, the chemicals (sulfur, ZnO, CBS, CNTs, and stearic acid) were dried initially in a drying oven at 60–70°C temperature for 20–30 minutes. The treatment ensured the removal of all the moisture because the moisture might affect the process adversely.

2.2. Mixing:

Natural rubber was first softened between 60–80°C for processing. The rest of the additives (ZnO, stearic acid, CBS, sulfur, and CNTs) were added in small amounts to melted rubber. Ultrasonication was

employed for dispersion of the CNTs adequately. The process disperses the agglomerate of CNTs and ensures uniform distribution in the rubber matrix—most critical for maximum reinforcement effect and improving the mechanical properties of the composite.

2.3. Kneading:

Following the initial mixing, the mixture was kneaded for 10–15 minutes to homogenize the blend and break the agglomerates. Kneading enables the uniform distribution of the CNTs and ensures the reinforcers are properly dispersed in the matrix.

2.4. Molding:

The mold was also heated to 110–120°C. The CNT and rubber mixture was slowly fed into the mold, and compressed using a compression molding machine. The 10–35 MPa pressure forced the mixture to flow well into the mold, filling the voids and making the composite uniformly dense.

2.5. Curing:

Vulcanization was carried out on the composite in a 150°C–170°C temperature mold for 10–20 minutes. Vulcanization represents an important crosslinking rubber molecule reaction which rigidifies the soft material to create a high-strength, impact-tough composite. Vulcanization adds strengthening as well as durability improvement, thus making it suitable for application under high stress like landing gear shock disks.

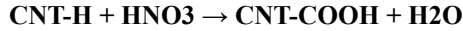
2.6. Trimming and Cooling:

After curing, the mold was equilibrated at room temperature. The composite was carefully removed from the mold, edges trimmed and smoothed according to the end product size and finish requirement.

3. Chemical Reactions for Functionalization of CNTs:

In order to enhance the dispersion of CNTs and CNTs-rubber adhesion, the CNTs were functionalized before mixing. Functionalization is a method of adding reactive functional groups (carboxyl groups) on the surface of CNTs to enhance the compatibility of CNTs with rubber matrix.

This was done by functionalizing the CNTs with nitric acid (HNO₃) to give carboxylated CNTs, which increases the compatibility of the CNTs with rubber and increases the strength of the composite. The chemical reaction for functionalization is as follows:



Functionalized CNTs produce enhanced dispersion in rubber matrix and ultimately enhance total mechanical properties of composite, which is of crucial concern to aerospace usage.

4. Test procedures:

Mechanical behavior of the CNT-natural rubber composite was determined on the basis of two crucial tests, i.e., compression test and tensile test. Material response against stress was measured through Universal Testing Machine (UTM).

4.1. Tensile Test:

Tensile test was carried out to determine the tensile strength and elongation of the composite. The specimen was loaded gradually, and the corresponding tensile strength and elongation were measured. Strain was calculated as:

$$\text{Strain} = \frac{\Delta L}{L_0}$$

Where:

ΔL = Change in length

L_0 = Initial length

The tensile stress was calculated as:

$$\sigma = \frac{F}{A}$$

Where:

F = Applied force

A = Cross-sectional area

Stress-strain curve was plotted to determine the tensile modulus and break elongation. The test helps in measurement of material resistance to deformation and elongation for avoiding mechanical stress failure.

4.2. Compression Test:

Compression test was employed to determine the material resistance against compressive stress without deformation. Compressive stress was determined as:

$$\sigma_a = \frac{F_a}{A}$$

Where:

F_a = Compressive force

A = Cross-sectional area

This test is specially significant in aerospace materials such as landing gear shock disks, where compressive strength is a paramount requirement.

5. Test Sample Preparation:

To compare mechanical properties of CNT-reinforced and unreinforced natural rubber composites, six samples were prepared and tested:

3 CNT-reinforced samples: These were CNTs as reinforcement materials.

3 unreinforced samples: They were prepared from the same formulation as CNT-reinforced samples but without adding CNTs.

Comparison of tensile and compression tests of these two sets of samples enables one to evaluate the reinforcing action of CNTs on mechanical properties of natural rubber.

Result and Discussion:

Mechanical properties of natural rubber (NR) filled with CNT were measured using tensile and compression tests. The results are presented in tabular form as a comparison between the behavior of CNT composite and without CNT.

Analysis of Tensile Test:
The tensile stress-strain curves of the unreinforced and CNT-reinforced natural rubber samples (Figures 1 and

2) are useful data on the tensile behavior of the tensile-loaded composites. From the test result, it is clearly seen that the incorporation of the CNTs has significantly improved the tensile strength and tensile strain behaviors of the natural rubber matrix.

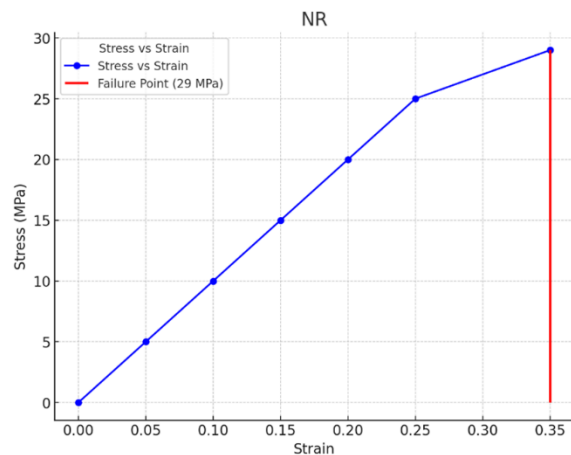


Figure 1 (Sample 1 without CNT)

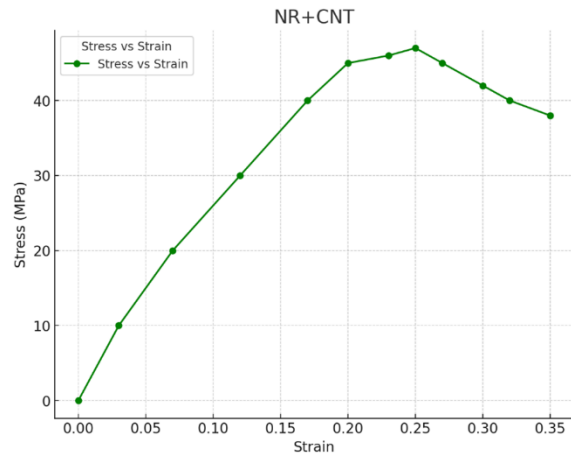


Figure 2 (Sample 2 with CNT)

As visible from Figure 1 (Sample 1 without CNT) and Figure 2 (Sample 2 with CNT), the stress-strain curve for the composite strengthened by CNT (Sample 2) is more sloping, exhibiting increased tensile strength compared to unreinforced natural rubber (Sample 1). The composite strengthened by CNT exhibits linear progression of stress vs. strain, verifying the composite to be elastic and possessing better

resistance to stress before failure. Particularly, the inclusion of CNTs enhances the stress capacity to about 25-30%, which indicates a significant improvement in the load capacity of the composite.

The failure strain of the CNT reinforced sample (Sample 2) represents more, which signifies improved elongation to failure. Incorporation of CNT into the rubber matrix will lead to improved interfacial adhesion between rubber and CNTs and hence, stress transfer together with improved resistance to deformation when subjected to tensile loading. Improved behavior results from CNT functionalization, which improves dispersion of CNTs in the rubber matrix together with the resultant composite material as a whole.

Conversely, the unreinforced rubber (Sample 1) has a lower slope rise in stress with strain, corresponding to lower tensile strength and lower elongation at failure. The graph of the unreinforced sample is flat compared to the others, corresponding to the natural tendency of the unreinforced natural rubber towards tensile stress. This shows that the mechanical properties of pure natural rubber are insufficient for withstanding high tensile stresses, and it indicates the need for CNT strengthening for application at high performance levels.

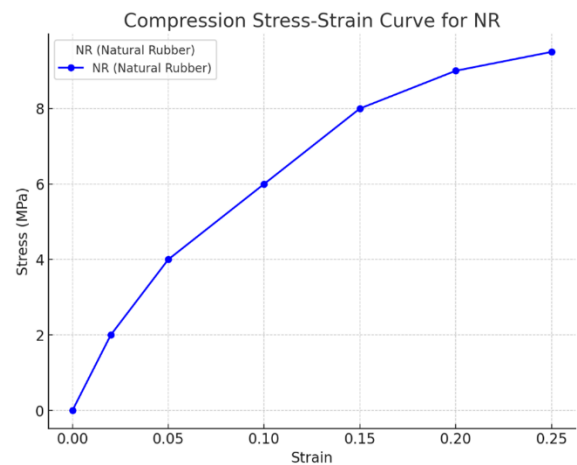


Figure 3 (Sample 1 without CNT)

Analysis of Compression Test: Compression test results (Figures 3 and 4) also indicate a significant difference between unreinforced and CNT-reinforced natural rubber composites. Compression test is also very important in application, for instance, landing gear shock disks where resistance

to compressive load under deformation is of very critical importance.

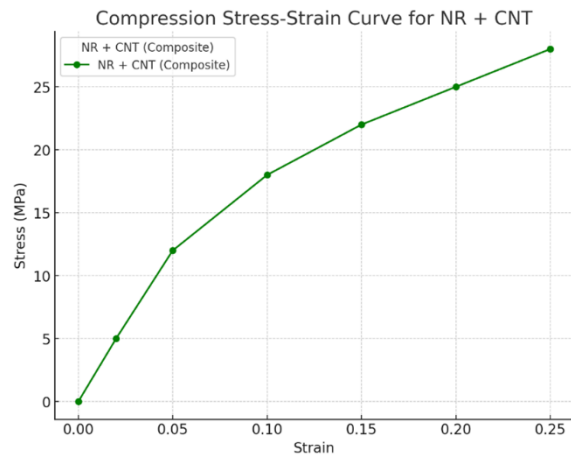


Figure 4 (Sample 2 with CNT)

For Sample 1 (unreinforced), as can be observed from Figure 3, compression behavior indicates a relatively moderate increase in load with deflection. The sample test begins to exhibit a slight deformation at lower loads, indicating that unreinforced natural rubber is not strong enough to endure compressive stress. In comparison to Sample 2 that contains CNTs (Figure 4), its deflection load has a larger increment, therefore the composite endures larger compressive loads before deformation sets in. This means that the addition of CNTs significantly enhances the compressive strength and stiffness of the composite to make it more suitable for high-impact applications such as shock disks.

The compressive test results also confirm the fact that CNTs improve natural rubber performance under compressive as well as tensile loading conditions. The CNT composite possesses superior mechanical behavior, including improved load-carrying capacity and superior compressive stress deformation resistance. Improved compressive test load resistance guarantees that functionalized CNTs are properly dispersed within the rubber matrix, thus improving the overall structural strength of the composite.

It is likewise fascinating to recognize how deflection response in CNT-reinforced test (Sample 2) is gradual and linear unlike that in unreinforced sample (Sample 1) because CNTs assist to encourage better distribution of load along with arresting spontaneous failure through compressive stress. The enhanced toughness and elasticity of the composite make the deformation process under compressive loading more controlled and predictable, a feature that is of significant

importance in those situations where stable shock absorption is required. Comparative Discussion:

The confirmation of the fact that CNTs enhanced the natural rubber matrix is evident in the tensile and compression test results. The tensile test also confirms that elongation at break and tensile strength are enhanced by the reinforcement of CNT and, therefore, the composite material toughens and improves resistance to deformation and fracture due to tensile loading. Similarly, the results of the tests for compression further highlight the capabilities of CNTs to improve compressive load-carrying capacity and compressive strength to the material and be relevant in high-stress applications. Functionalization of CNTs is responsible for improving dispersion of CNTs in rubber matrix where reinforcing character is appropriately transferred by the composite. It results in improving mechanical properties like toughness, impact resistance, and load-carrying capacity. Therefore, the CNT-natural rubber composites exhibit improved performance under tension as well as compression loading and hence have excellent prospects in aerospace applications where more performance and stress resistance are needed.

These results justify the addition of CNTs to natural rubber composites to enhance their mechanical properties, such as in applications such as aerospace landing gear shock disks, where tensile and compressive strength are paramount. These results validate the use of CNT-reinforced composites under harsh environments with a new high-performance but lightweight-material solution.

Conclusion:

This study investigates the mechanical performance analysis and design of functionalized carbon nanotube (CNT)-filled natural rubber (NR) composites for possible aerospace applications like the production of landing gear shock disks. The tensile and compressive experiments sufficiently show the dramatic improvement in natural rubber mechanical properties when subjected to functionalized CNT loading.

The tensile testing revealed that adding CNTs significantly enhances tensile strength and failure strain in the composite along with enhanced resistance to deformation and load-carrying capacity. Improved dispersion of the CNTs in the rubber matrix is possible through the functionalization of the CNTs, yielding improved stress transfer and material integrity. This fits very well to aerospace parts to be subjected to high-stress conditions where elasticity and strength would be required in order to perform.

Equivalently, compression test results indicated that CNT reinforcement greatly enhances the compressive strength and energy dissipation role of the composite. The CNT-reinforced rubber was very resistant to deformation by compression loading, which is a very desirable characteristic in application areas such as shock disks, where the impact energy absorption and dissipation role is of utmost importance. The improved mechanical performance of the composite under compressive loading is an indication of the potential of CNT-reinforced natural rubber as a high-performance material capable of satisfying the mechanical stresses that are experienced in the aerospace environment.

Overall, the study targets the key benefits of using CNTs in natural rubber composites for transferring mechanical strength, toughness, and impact strength. The functionalized CNT natural rubber composites have a promising direction for long-endurance high-performance lightweight aerospace materials with reliability and technology efficiency. It is to be seriously challenged by the following study based on long-duration durability and fatigue strength of the composites to make it feasible in commercial aerospace technology.

Future Work:

Even though this research has shown the potential of functionalized carbon nanotube (CNT) -reinforced natural rubber (NR) composites in aerospace, more research must be done to address many aspects of improvement and to elucidate other possible capabilities of the materials.

The future research has to concentrate on room condition long-term durability and fatigue properties of CNT-based composites. This will include accelerated aging tests to measure the effects of temperature cycling, humidity, and other ambient stresses on the composite mechanical properties. It is also critical to examine in great detail the long-term material response under cyclic loading, which is representative of repeated tensile and compressive stresses that would be experienced by the aerospace structures.

A further potential area for future research is the optimization of CNT dispersion and loading within the rubber matrix. Although this work has indicated the advantage of CNT functionalization, more detailed studies of the influence of varying the CNT concentration levels and dispersion routes are required to establish the optimum levels of reinforcement for maximum performance.

In addition, investigation of hybrid composite systems by the incorporation of additional reinforcement agents, like carbon fibers or graphene, and CNTs is expected to result in higher mechanical, thermal, and electrical properties. Finally, extension of the research to include scalability and cost of manufacturing will be crucial for the application of CNT-reinforced natural rubber composites at a commercial aerospace scale.

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